N. Robertson and P. D. Seymour, Graph Minors. I. Excluding a Forest, *J. Combin. Theory Ser. B* **35** (1983), 39–61.

## About this copy

This is a corrected copy of the paper cited above. The text is retyped from the printed paper. It keeps the paper's wording and the numbering of its results, equations and footnotes, and a marker such as [p. $N$] shows where page $N$ of the printed paper begins.

The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM01.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages:

1. **Missing proof.** Proof of (1.2), page 40: the derivation from (1.1), with the proofs of the observations (a) and (b), is added.
2. **Missing proof.** Proofs of (1.4)–(1.6), page 40: the proofs are added.
3. **Misprint.** Statement of (1.7), page 41: $1 < i, i' \leq T$ becomes $1 \leq i, i' \leq T$.
4. **Misprint.** Proof of (1.7), page 42: $\rho$ becomes $e$.
5. **Degenerate case.** Statement of (2.4), page 44: the hypothesis $m \geq \frac{1}{2}\theta^2$ is added.
6. **Degenerate case.** Statement of (2.6), page 46: the hypothesis $m > 0$ becomes $m \geq \frac{1}{2}\theta^2$.
7. **Misprint.** Proof of (2.7), page 48: $v_0$ becomes $v$.

8. **Degenerate case.** Statement of (2.8), page 48: the hypothesis $d+r > 0$ is added.
9. **Misprint.** Statement of (3.2), page 51: the exponent $(\lambda+1)/2$ becomes $\lambda/2+1$.
10. **Missing proof.** Proof of (3.2), page 51: the induction is added.
11. **Missing proof.** Proof of (3.3), page 51: the proof is added.
12. **Missing proof.** Proof of (3.6), page 53: the proof of the four inequalities is added.
13. **Misprint.** Statement of (4.4)(i), page 56: $j \geq 1$ becomes $j \geq i$.
14. **Misprint.** Proof of (4.14), page 59: an extra closing bracket after $\gamma - 1$ is removed.

[p. 39] # Graph Minors. I. Excluding a Forest

NEIL ROBERTSON AND P. D. SEYMOUR*
*Department of Mathematics, Ohio State University,*
*231 W. 18th Ave, Columbus, Ohio 43210*
*Communicated by the Editors*
Received May 3, 1982

The path-width of a graph is the minimum value of $k$ such that the graph can be obtained from a sequence of graphs $G_1, ..., G_r$ each of which has at most $k+1$ vertices, by identifying some vertices of $G_i$ pairwise with some of $G_{i+1}$ $(1 \leq i < r)$. For every forest $H$ it is proved that there is a number $k$ such that every graph with no minor isomorphic to $H$ has path-width $\leq k$. This, together with results of other papers, yields a "good" algorithm to test for the presence of any fixed forest as a minor, and implies that if $P$ is any property of graphs such that some forest does not have property $P$, then the set of minor-minimal graphs without property $P$ is finite.

## 1. INTRODUCTION

Let $G$ be a graph. (All graphs in this paper are finite, and may have loops or multiple edges unless we state otherwise.) A sequence $X_1, ..., X_r$ of subsets of $V(G)$ (the vertex set of $G$) is a *path-decomposition* of $G$ if the following conditions are satisfied.

(W1) *For every edge $e$ of $G$, some $X_i$ $(1 \leq i \leq r)$ contains both ends of $e$.*

(W2) *For $1 \leq i \leq i' \leq i'' \leq r$, $X_i \cap X_{i''} \subseteq X_{i'}$.*

The *path-width* of $G$ is the minimum value of $k \geq 0$ such that $G$ has a path-decomposition $X_1, ..., X_r$ with $|X_i| \leq k+1$ $(1 \leq i \leq r)$.

$H$ is a *minor* of $G$ if $H$ can be obtained from $G$ by deleting some vertices and/or edges, and/or contracting some edges. The main theorem of this paper is the following:

(1.1). *For every forest $H$ there is an integer $w$ such that every graph with no minor isomorphic to $H$ has path-width $\leq w$.*

[p. 40] There are three attractive features of the theorem which serve as its motivation. First, the theorem is, in a sense, sharp, for it can be reformulated as follows:

*Partially supported by National Science Foundation Grant MCS 8103440.

(1.2). *Let $\mathscr{F}$ be a set of graphs. Then the following are equivalent*:

(i) *for some integer $w$, every $G \in \mathscr{F}$ has path-width $\leq w$*;

(ii) *there is a forest $H$ such that no $G \in \mathscr{F}$ has a minor isomorphic to $H$.*

It is easy to prove the equivalence of (1.2) with (1.1) by making use of two observations: (a) if $H$ is a minor of $G$, then its path-width is no greater than the path-width of $G$, and (b) there are trees with arbitrarily large path-width (e.g., the tree $\mathbf{Y}_\lambda$ defined later has path-width $\lceil \frac{1}{2}(\lambda+1)\rceil$, for any integer $\lambda \geq 1$, where $\lceil x\rceil$ is the least integer not less than $x$).

We give the details. For (a), let $H$ be isomorphic to a minor of $G$. Then $H$ is isomorphic to a graph obtained from a subgraph $G'$ of $G$ by contracting a set $C$ of edges. For each $v \in V(H)$ let $Y(v)$ be the vertex set of the component of the graph with vertex set $V(G')$ and edge set $C$ that is contracted to $v$, and for each edge $e$ of $H$ let $f(e)$ be the corresponding edge of $G'$. Then the sets $Y(v)$ ($v \in V(H)$) are disjoint, each $G|Y(v)$ is connected, and for each edge $e$ of $H$ with ends $v, v'$, the edge $f(e)$ has one end in $Y(v)$ and the other in $Y(v')$. (These are the conditions (M1)–(M3) of Section 3.) We use the following.

(1) *Let $X_1, \ldots, X_r$ be a path-decomposition of $G$, let $K$ be a connected subgraph of $G$, and let $1 \leq i \leq j \leq l \leq r$. If $K$ has a vertex in $X_i$ and a vertex in $X_l$, then $K$ has a vertex in $X_j$.*

Suppose not, and let $x \in V(K) \cap X_i$ and $y \in V(K) \cap X_l$. We claim that every vertex of $K$ is in some $X_t$ with $t \leq j$. This holds for $x$, since $i \leq j$. Let $z, w$ be adjacent vertices of $K$, where $z \in X_t$ for some $t \leq j$. By (W1) some $X_s$ contains both $z$ and $w$. If $s > j$, then $z \in X_t \cap X_s \subseteq X_j$ by (W2), which is impossible since $K$ has no vertex in $X_j$. So $s \leq j$, and $w \in X_s$. Since $K$ is connected, every vertex of $K$ is reached from $x$ by such steps, and the claim follows. In particular $y \in X_t$ for some $t \leq j$, and so $y \in X_t \cap X_l \subseteq X_j$ by (W2), which is impossible. This proves (1).

Now let $X_1, \ldots, X_r$ be a path-decomposition of $G$ with $|X_i| \leq k+1$ ($1 \leq i \leq r$). For $1 \leq i \leq r$ let $X'_i$ be the set of all $v \in V(H)$ with $Y(v) \cap X_i \neq \varnothing$. We claim that $X'_1, \ldots, X'_r$ is a path-decomposition of $H$ with $|X'_i| \leq k+1$ ($1 \leq i \leq r$). For (W1), let $e$ be an edge of $H$ with ends $v, v'$. Then $f(e)$ has an end $a \in Y(v)$ and an end $b \in Y(v')$, and by (W1) for $G$ some $X_i$ contains $a$ and $b$. So $v, v' \in X'_i$. For (W2), let $1 \leq i \leq j \leq l \leq r$ and $v \in X'_i \cap X'_l$. Then $G|Y(v)$ is a connected subgraph of $G$ with a vertex in $X_i$ and a vertex in $X_l$, and so it has a vertex in $X_j$, by (1). Thus $v \in X'_j$. For the sizes, choose for each $v \in X'_i$ a vertex in $Y(v) \cap X_i$. Since the sets $Y(v)$ are disjoint, distinct members of $X'_i$ give distinct vertices of $X_i$, and so $|X'_i| \leq |X_i| \leq k+1$. This proves (a).

For (b) we prove that $\mathbf{Y}_{2k+1}$ has path-width $> k$ for every integer $k \geq 0$. (This is the lower bound above for odd $\lambda$, which is all that we need.) For $k = 0$ it is true, since $\mathbf{Y}_1$ has an edge, and by (W1) some set of any path-decomposition contains both of its ends. Let $k \geq 0$, suppose that $\mathbf{Y}_{2k+1}$ has path-width $> k$, and let $\mathbf{Y} = \mathbf{Y}_{2k+3}$. Let $c$ be the vertex of valency 3 of the copy of $K_{1,3}$ from which $\mathbf{Y}$ is built, and let $K_1, K_2, K_3$ be the components of $\mathbf{Y}\backslash\{c\}$. Each $K_j$ is isomorphic to the tree $\mathbf{H}_{2k+2}$ of Section 3, and so it has a minor isomorphic to $\mathbf{Y}_{2k+1}$, by (3.3). Hence $K_j$ has path-width $> k$, by (a). Now suppose that $X_1, \ldots, X_r$ is a path-decomposition of $\mathbf{Y}$ with $|X_i| \leq k+2$ $(1 \leq i \leq r)$. For $j = 1, 2, 3$ the sets $X_i \cap V(K_j)$ $(1 \leq i \leq r)$ form a path-decomposition of $K_j$, and so $|X_{i_j} \cap V(K_j)| \geq k+2$ for some $i_j$. Then $X_{i_j} \subseteq V(K_j)$ and $X_{i_j} \neq \varnothing$. Since $K_1, K_2, K_3$ are disjoint, $i_1, i_2, i_3$ are distinct, and we may assume that $i_1 < i_2 < i_3$. The graph $\mathbf{Y}\backslash V(K_2)$ is connected, since $c$ is adjacent to a vertex of $K_1$ and to a vertex of $K_3$. It contains $K_1$ and $K_3$, and so it has a vertex in $X_{i_1}$ and a vertex in $X_{i_3}$. By (1) it has a vertex in $X_{i_2}$. But $X_{i_2} \subseteq V(K_2)$, which is impossible. Thus $\mathbf{Y}_{2k+3}$ has path-width $> k+1$, and (b) follows by induction.

Now assume (1.1), and let $\mathscr{F}$ be a set of graphs. If (i) holds, let $H = \mathbf{Y}_{2w+1}$, which is a tree. If some $G \in \mathscr{F}$ had a minor isomorphic to $H$, then $H$ would have path-width $\leq w$ by (a), contrary to (b). So (ii) holds. If (ii) holds, (1.1) applied to $H$ gives (i).[F1]

The second attractive feature of (1.1) is that it dovetails nicely with the main theorem of [3]; the two results together yield the following:

(1.3). *Let $G_1, G_2, \ldots$, be a countable sequence of graphs, such that $G_1$ is a forest. Then there exist $j > i \geq 1$ such that $G_i$ is isomorphic to a minor of $G_j$.*

(This implies the final result mentioned in the abstract, as may easily be seen.) Theorem (1.3) is relevant to an interesting conjecture of Wagner (unpublished), that (1.3) is true even without the hypothesis that $G_1$ is a forest. See [3] for a more complete discussion.

And third (see [1]), (1.1) yields a "good" algorithm to test if an arbitrary graph $G$ has a minor isomorphic to a fixed forest $H$. ("Good" here is in its technical sense of "polynomially bounded." It is not a practical algorithm; the exponent of the polynomial, although constant, is enormous.)

For the purposes of this paper we need the following three facts about path-width (the proofs are easy).

[F1] **Missing proof.** The paper leaves the derivation of (1.2) from (1.1), and the proofs of the observations (a) and (b), to the reader. The argument above is the one used in the formalization. It uses (3.3), whose proof does not depend on (1.2).

(1.4). *If every connected component of $G$ has path-width $\leq k$, then $G$ has path-width $\leq k$.*

(1.5). *If $X \subseteq V(G)$ and $G\backslash X$ has path-width $\leq k$, then $G$ has path-width $\leq k + |X|$.*

($G\backslash X$ denotes the result of deleting $X$.)

(1.6). *If $X_1, ..., X_r \subseteq V(G)$ is a path-decomposition of $G$, and for $1 \leq i \leq r-1$, $|X_i \cap X_{i+1}| \leq k$, and for $1 \leq i \leq r$,*

$$G\backslash \bigcup (X_j : 1 \leq j \leq r, j \neq i)$$

*has path-width $\leq k'$, then $G$ has path-width $\leq k' + 2k$.*

We give the proofs. First, if $G$ has path-width $\leq k$, then it has a path-decomposition $X_1, \ldots, X_r$ with $|X_i| \leq k+1$ $(1 \leq i \leq r)$ in which every vertex is in some $X_i$. For by (W1) a vertex in no $X_i$ is incident with no edge, and we may add the set $\{v\}$ at the end of the sequence for each such vertex $v$. Then (W1) and (W2) still hold.

For (1.4), take a path-decomposition of each component of $G$ with sets of size $\leq k+1$, and write these sequences one after another. The result is a path-decomposition of $G$. For every edge of $G$ is an edge of some component, and each vertex is in sets of only one of the sequences, since the sets of each sequence are subsets of the vertex set of its component.

For (1.5), take a path-decomposition $X_1, \ldots, X_r$ of $G\backslash X$ with $|X_i| \leq k+1$ $(1 \leq i \leq r)$ in which every vertex of $G\backslash X$ is in some $X_i$. We claim that $X_1 \cup X, \ldots, X_r \cup X, X$ is a path-decomposition of $G$. Its sets have size $\leq k+1+|X|$, and so this will prove (1.5). For (W1), every set contains $X$, and so an edge with both ends in $X$ has both ends in the last set. An edge with just one end in $X$ has its other end in some $X_i$, and so both ends in $X_i \cup X$. An edge with no end in $X$ is an edge of $G\backslash X$, and so both its ends are in some $X_i$. For (W2), a vertex in $X$ is in every set, and a vertex not in $X$ is in $X_i \cup X$ exactly when it is in $X_i$, and is not in the last set.

For (1.6), let $X_1, \ldots, X_r$ be as in (1.6). For $1 \leq i \leq r$ let $B_i = X_i \cap \bigcup (X_j : j \neq i)$ and $G_i = G\backslash \bigcup (X_j : j \neq i)$. A vertex in $X_i \cap X_j$ with $j < i$ is in $X_{i-1}$, and one with $j > i$ is in $X_{i+1}$, by (W2). So $B_i \subseteq (X_{i-1} \cap X_i) \cup (X_i \cap X_{i+1})$ (where $X_0 = X_{r+1} = \varnothing$), and $|B_i| \leq 2k$. By hypothesis and the first paragraph, $G_i$ has a path-decomposition $Z_{i1}, \ldots, Z_{im_i}$ with $|Z_{ij}| \leq k'+1$ $(1 \leq j \leq m_i)$ in which every vertex of $G_i$ is in some set. We call the sequence

$$(Z_{i1} \cap X_i) \cup B_i,\ \ldots,\ (Z_{im_i} \cap X_i) \cup B_i,\ B_i$$

the *refinement* of $X_i$. Its sets are subsets of $X_i$ of size $\leq k' + 1 + 2k$, and each of them contains $B_i$.

(2) *For $1 \le i \le r$, the refinement of $X_i$ is a path-decomposition of $G|X_i$, and every vertex of $X_i$ is in some set of it.*

A vertex of $X_i - B_i$ is in no $X_j$ with $j \neq i$, and so it is a vertex of $G_i$, and it is in some $Z_{ij}$. The vertices of $B_i$ are in the last set. For (W1), let $e$ be an edge of $G|X_i$ with ends $u, v$. If $u, v \notin B_i$, then $u, v \notin \bigcup(X_j : j \neq i)$, so $e$ is an edge of $G_i$, and some $Z_{ij}$ contains $u$ and $v$, by (W1) for $G_i$. Then $(Z_{ij} \cap X_i) \cup B_i$ contains both. If $u \in B_i$, then $v$ is in some set of the refinement, as we just saw, and $u$ is in every set. The case $v \in B_i$ is the same. For (W2), a vertex of $B_i$ is in every set, and a vertex $v \in X_i - B_i$ is in $(Z_{ij} \cap X_i) \cup B_i$ exactly when $v \in Z_{ij}$, and is not in the last set. So (W2) follows from (W2) for $Z_{i1}, \ldots, Z_{im_i}$. This proves (2).

Now write the refinements of $X_1, \ldots, X_r$ one after another. We claim that the result is a path-decomposition of $G$. Since its sets have size $\le k' + 1 + 2k$, this will prove (1.6). For (W1), let $e$ be an edge with ends $u, v$. By (W1), $u, v \in X_i$ for some $i$. So $e$ is an edge of $G|X_i$, and some set of the refinement of $X_i$ contains $u$ and $v$, by (2). For (W2), let $v$ be in a set of the refinement of $X_i$ and in a later set of the refinement of $X_l$, where $i \le l$, and consider a set between them, in the refinement of $X_j$ with $i \le j \le l$. If $i = l$, then $j = i$, and $v$ is in the set by (2). Suppose that $i < l$. The sets of the refinements of $X_i$ and $X_l$ are subsets of $X_i$ and $X_l$, and so $v \in X_i \cap X_l$, and $v \in X_j$ by (W2). Since $i < l$, one of $i, l$ is not $j$, and so $v$ is in $X_j$ and in some other set $X_{j'}$. Hence $v \in B_j$. So $v$ is in every set of the refinement of $X_j$.[F2]

[p. 41] Our proof of (1.1) is basically by induction on the "complexity" of the forest $H$. This is explained in detail in Section 3, and the induction argument is performed in Sections 4 and 5. Section 2 contains some crucial lemmas about "grids."

Our terminology is mostly standard, but a few terms need explanation. If $X, Y$ are sets, $X - Y$ denotes

$$\{x : x \in X, x \notin Y\}.$$

If $G$ is a graph, $V(G)$ and $E(G)$ denote its vertex-set and edge-set, respectively. A graph is *simple* if it has no loops or multiple edges. If $X \subseteq V(G)$, $G\backslash X$ denotes the result of deleting from $G$ all the vertices in $X$, and $G \,|\, X$ denotes $G\backslash(V(G) - X)$. A graph $G$ is a *subdivision* of a graph $H$ if $H$ can be obtained from $G$ by (repeatedly) choosing a vertex of valency 2 and contracting an edge incident with it which is not a loop. Two subgraphs of a

[F2] **Missing proof.** The paper says that the proofs of (1.4)–(1.6) are easy and does not give them. The arguments above are the ones used in the formalization.

graph are *disjoint* if they have no common vertices, and a set of subgraphs is disjoint if every pair of its members are disjoint. Every *path* has at least one vertex, and no "repeated" vertices. It has an *initial* and a *terminal* vertex (which are equal for a one-vertex path) called its *ends*, and any other vertices are called *internal* vertices. A path *avoids* $X \subseteq V(G)$ if it has no vertex in $X$. $X \subseteq V(G)$ *separates* $Y, Z \subseteq V(G)$ if no path in $G$ from $Y$ to $Z$ avoids $X$. (Thus $Y \cap Z \subseteq X$.) We say that $X$ separates $y, z \in V(G)$ if it separates $\{y\}, \{z\}$.

We shall require the following lemma:

(1.7). *Let $\rho$ be a vertex of a graph $G$, and let $Y_1, Y_2, ..., Y_T$ be disjoint subsets of $V(G)$, where $T \geq 1$. Suppose that for $1 \leq i, i' \leq T$,*[F3] *and for all $y \in Y_{i'}$, $Y_i$ separates $y$ and $\rho$ if and only if $i \leq i'$. For $1 \leq i < T$, let $W_i$ be the set of all vertices of $G$ separated from $\rho$ by $Y_i$, and not separated from $\rho$ by $Y_{i+1}$. Let $W_0$ be the set of vertices not separated from $\rho$ by $Y_1$, and let $W_T$ be the set of all vertices separated from $\rho$ by $Y_T$. Then the sets $W_0, W_1, ..., W_T$ are disjoint and partition $V(G)$, and $Y_i \subseteq W_i$ $(1 \leq i \leq T)$, and the sequence*

$$W_0 \cup Y_1, W_1 \cup Y_2, ..., W_{T-1} \cup Y_T, W_T$$

*is a path-decomposition of $G$.*

*Proof.* For every vertex $v \in V(G)$, there is a unique value of $i$ with $0 \leq i \leq T$ such that

(i) either $i = 0$ or $Y_i$ separates $v$ and $\rho$, and

(ii) either $i = T$ or $Y_{i+1}$ does not separate $v$ and $\rho$.

For it is clear that there is at least one such value of $i$; and if $i, i'$ are two [p. 42] values, with $i < i'$ say, then $i < T$ and $i' > 0$, and we have that $Y_{i+1}$ does not separate $v$ and $\rho$, and $Y_{i'}$ separates $v$ and $\rho$. Let $P$ be a path in $G$ from $v$ to $\rho$ avoiding $Y_{i+1}$. Then $P$ meets $Y_{i'}$, and so $Y_{i+1}$ does not separate $Y_{i'}$ and $\rho$, contrary to our hypothesis.

It follows that for every vertex $v$ there is a unique value of $i$ with $v \in W_i$, and so $W_0, W_1, ..., W_T$ partition $V(G)$. Clearly, $Y_i \subseteq W_i$ $(1 \leq i \leq T)$. It remains to prove that the sequence

$$W_0 \cup Y_1, W_1 \cup Y_2, ..., W_{T-1} \cup Y_T, W_T$$

[F3] **Misprint.** As printed, nothing is assumed about $Y_1$, and the conclusion fails. For the path $\rho\, a\, b$ with $T = 2$, $Y_1 = \{b\}$ and $Y_2 = \{a\}$, the hypothesis holds, but $W_0 = \{\rho, a\}$ meets $W_2 = \{a, b\}$. The proof uses the case $i = 1$ (in the uniqueness step, $Y_{i+1}$ with $i = 0$ is $Y_1$), and the use in Section 4, through (4.5), has $1 \leq i, i' \leq T$.

is a path-decomposition of $G$. We need only show (W1), since (W2) is obvious.

Let $e$ be an edge of $G$, with ends $u, v$ say. Let $u \in W_i$, $v \in W_j$, where $i \le j$ say. If $i = j$, then $\rho e$[F4] has both ends in some set in the sequence, as required; we assume therefore that $i < j$. Thus $Y_{i+1}$ separates $v$ and $\rho$. It does not separate $u$ and $\rho$, since $u \in W_i$, and yet $u, v$ are adjacent. It follows that $v \in Y_{i+1}$, and so $e$ has both ends in $W_i \cup Y_{i+1}$, as required.

## 2. Grids

If $\theta \ge 2$, the *$\theta$-grid* is the simple graph with vertices $v_{ij}$ $(1 \le i, j \le \theta)$ in which $v_{ij}$ and $v_{i'j'}$ are adjacent if $|i - i'| + |j - j'| = 1$. $\mathscr{F}_\theta$ denotes the set of all graphs with no minor isomorphic to the $\theta$-grid. For the remainder of the paper, $\theta \ge 6$ will be fixed and, for convenience, even. In this section we prove some connectivity results about the graphs in $\mathscr{F}_\theta$.

(2.1). *If $P_1, ..., P_m$ are disjoint paths of a graph $G \in \mathscr{F}_\theta$, and $B_1, ..., B_n$ are also disjoint paths, and each $P_i$ has exactly one vertex in common with each $B_j$, and each $P_i$ meets $B_1, ..., B_n$ in that order, then either $m < \theta$ or $n < \frac{1}{2}\theta \cdot \theta!$.*

*Proof.* We assume that $m \ge \theta$. Since there are only $\frac{1}{2}(\theta!)$ different orderings of $P_1, ..., P_\theta$ up to reversal, it follows that there exists $J \subseteq \{1, ..., n\}$ with $|J| \ge n/\frac{1}{2}(\theta!)$ such that for $j, j' \in J$, $B_j$ and $B_{j'}$ meet $P_1, ..., P_\theta$ in the same order or the reverse. If $n \ge \frac{1}{2}\theta \cdot \theta!$, then $|J| \ge \theta$ and so $G$ has a minor isomorphic to the $\theta$-grid, which is impossible. Thus $n < \frac{1}{2}\theta \cdot \theta!$.

(2.2). *If $P_1, ..., P_m$ are disjoint paths of $G \in \mathscr{F}_\theta$, and $B_1, ..., B_n$ are disjoint stars, and each $P_i$ has exactly one vertex in common with each $B_j$, and no $P_i$ contains the centre of any star $B_j$, and each $P_i$ meets $B_1, ..., B_n$ in that order, then either $m < \theta$ or $n < \theta(\theta - 1)$.*

(A *star* is a tree in which one vertex, called the *centre*, is adjacent to all others.)

[p. 43] *Proof.* We may assume, by deletion, that every vertex and edge of $G$ is either in some $P_i$ or in some $B_j$; and by contraction, that every vertex is in some $B_j$. For $1 \le j \le n$ let $v(j)$ be the centre of the star $B_j$, and let the common vertex of $P_i$ and $B_j$ be $v(i, j)$ $(1 \le i \le m)$. We assume, for a contradiction, that $m \ge \theta$ and $n \ge \theta(\theta - 1)$. For $1 \le k \le \theta$, let $Q_k$ be the

[F4] **Misprint.** The edge $e$ is meant.

subgraph of $G$ induced by

$$\bigcup_{1\le l\le\theta-1}\{v(l,(k-1)(\theta-1)+l),\\ v((k-1)(\theta-1)+l),v(l+1,(k-1)(\theta-1)+l)\}.$$

We see that for $1\le k\le\theta$, $Q_k$ is a path, meeting $P_1,...,P_\theta$ in that order, and each $P_i$ meets $Q_1,...,Q_\theta$ in that order. But then $G$ has a minor isomorphic to the $\theta$-grid, a contradiction.

(2.3). *If $P_1,...,P_m$ are disjoint paths of $G\in\mathscr{F}_\theta$, and $B_1,...,B_n$ are disjoint connected subgraphs of $G$, and each $P_i$ meets each $B_j$, and for $1\le j<j'\le n$ all vertices of $B_j$ on $P_i$ occur before all vertices of $B_{j'}$, then either $m<\frac{1}{2}\theta^2$ or $n<\theta^{2\theta-2}$.*

*Proof.* If possible, choose a counterexample with $|E(G)|+m$ minimum. Then evidently we have $m=\frac{1}{2}\theta^2$.

If some $P_i$ has more than one vertex in common with some $B_j$, we may contract the edges of $P_i$ between the two vertices of $B_j$ and produce a smaller counterexample, a contradiction. Thus each $P_i$ meets each $B_j$ in exactly one vertex, and in particular no edge is both in some $P_i$ and in some $B_j$. Each $B_j$ is a tree, for if some $B_j$ is not, we may delete an edge from it and produce a smaller counterexample. Every vertex $v$ of each $B_j$ is on some $P_i$; for if not, we may contract some edge of $B_j$ incident with $v$ and produce a smaller counterexample. It follows that each $B_j$ has exactly $\frac{1}{2}\theta^2$ vertices.

It is elementary that for $1\le j\le n$, $B_j$ has a subgraph $Q_j$ which is either a path with $\theta$ vertices or a tree with $\theta$ end-vertices.

For $1\le j\le n$, let $I_j\subseteq\{1,...,\frac{1}{2}\theta^2\}$ be $\{i\colon P_i$ meets $Q_j$ in some vertex which, if $Q_j$ is not a path, is an end-vertex of $Q_j\}$. Then $|I_j|=\theta$ for all $j$, and so there exists $J\subseteq\{1,...,n\}$ with

$$|J|\ge n\Big/\binom{\frac{1}{2}\theta^2}{\theta}$$

such that $I_j=I_{j'}$ for $j,j'\in J$.

Let $J_1\subseteq J$ be $\{j\in J\colon Q_j$ is a path$\}$ and let $J_2=J-J_1$. By (2.1), $|J_1|<\frac{1}{2}\theta\cdot\theta!$, and by (2.2) $|J_2|<\theta(\theta-1)$. Thus

$$|J|<\tfrac{1}{2}\theta\cdot\theta!+\theta(\theta-1)$$

[p. 44] and so

$$\begin{aligned}n&<\binom{\frac{1}{2}\theta^2}{\theta}(\tfrac{1}{2}\theta\cdot\theta!+\theta(\theta-1))\\&\le\theta^{2\theta-2}\end{aligned}$$

since $\theta \geq 6$, a contradiction, as required.

(2.4). *If $P_1, ..., P_m$ are disjoint paths of $G \in \mathscr{F}_\theta$, where $m \geq \frac{1}{2}\theta^2$, and $B_1, ..., B_n$ are disjoint connected subgraphs of $G$, and each $B_j$ meets at least $\frac{1}{2}\theta^2$ of $P_1, ..., P_m$, and for $1 \leq i \leq m$ and for $1 \leq j < j' \leq n$, all vertices of $B_j$ on $P_i$ occur before all vertices of $B_{j'}$, then $n < \theta^{2\theta-2}\binom{m}{\frac{1}{2}\theta^2}$.*[F5]

*Proof.* For $1 \leq j \leq n$, choose $I_j \subseteq \{1, ..., m\}$ with $|I_j| = \frac{1}{2}\theta^2$ such that $P_i$ meets $B_j$ for each $i \in I_j$. Now there exists $J \subseteq \{1, ..., n\}$ with $|J| \geq n\binom{m}{\frac{1}{2}\theta^2}^{-1}$, such that for $j, j' \in J$, $I_j = I_{j'} = I$ say. Then $|I| = \frac{1}{2}\theta^2$. Now each path $P_i$ ($i \in I$) meets each $B_j$ ($j \in J$) in order, and so by (2.3), $|J| < \theta^{2\theta-2}$. The result follows.

(2.5). *If $A_1, A_2 \subseteq V(G)$, and there is a unique set $\{P_1, ..., P_m\}$ of $m$ disjoint paths in $G$ from $A_1$ to $A_2$, and every vertex of $G$ is in one of these paths, then there is an integer-valued function $\mu$ defined on $V(G)$, such that*

(i) *if $v'$ is after $v$ on $P_i$, then $\mu(v') > \mu(v)$ $(1 \leq i \leq m)$;*

(ii) *for any integer $t$, $Z_t$ separates $\{v : \mu(v) < t\}$ from $\{v : \mu(v) \geq t\}$, where $Z_t = \{v :$ for some $i$, $v$ is the first vertex on $P_i$ with $\mu(v) \geq t\}$.*

*Proof. Step* 1. *There is no sequence $u_1, v_1, u_2, v_2, ..., u_r, v_r$ of distinct vertices of $G$, where $r \geq 1$, and for $1 \leq j \leq r$ $v_{j-1}, u_j$ are adjacent, joined by an edge not in any of $P_1, ..., P_m$ (setting $v_0 = v_r$), and $u_1, u_2, ..., u_r$ are all on different $P_i$'s, and for $1 \leq j \leq r$, $v_j$ occurs after $u_j$ on one of $P_1, ..., P_m$.*

For if there is such a sequence, let us renumber $P_1, ..., P_m$ for simplicity so that $u_i$ lies on $P_i$ $(1 \leq i \leq r)$. For $1 \leq i \leq r$, let $e_i$ be an edge of $G$ joining $v_{i-1}$ and $u_i$. Let $Q_i, R_i, S_i$ be the subpaths of $P_i$ from $A$ to $u_i$, from $u_i$ to $v_i$, and from $v_i$ to $B$, respectively. Then $R_i$ has at least one edge, since $u_i \neq v_i$. Setting $S_0 = S_r$, we have $Q_i$ and $S_{i-1}$ are disjoint $(1 \leq i \leq r)$ and so $Q_i$, $e_i$, and $S_{i-1}$ form a path $P'_i$ say from $A_1$ to $A_2$. Then $P'_1, ..., P'_r, P_{r+1}, ..., P_m$ are disjoint paths of $G$ from $A_1$ to $A_2$; and this set is different from $P_1, ..., P_m$, since none of $P_1, ..., P_m$ uses the edge $e_1$. This contradicts our hypothesis that $P_1, ..., P_m$ are unique.

*Step* 2. *There is no sequence $u_1, v_1, ..., u_r, v_r$ of vertices of $G$, where $r \geq 1$, such that for $1 \leq i \leq r$, $v_{i-1}, u_i$ are adjacent, joined by an edge not in any of $P_1, ..., P_m$ (setting $v_0 = v_r$), and for $1 \leq i \leq r$, $v_i$ occurs after $u_i$ on one of $P_1, ..., P_m$.*

[F5] **Degenerate case.** Without the added hypothesis the statement fails when $n = 0$ and $m < \frac{1}{2}\theta^2$, since then the binomial coefficient is 0 and the conclusion reads $0 < 0$ (for example, the empty graph with $m = 0$). If $n \geq 1$, then $B_1$ meets $\frac{1}{2}\theta^2$ of the paths, so $m \geq \frac{1}{2}\theta^2$ holds anyway. The only use, in the proof of (2.6), has $m \geq \frac{1}{2}\theta^2$ by the hypothesis added to (2.6).

[p. 45] If possible, choose such a sequence with $r$ minimum. If $u_1, u_2, ..., u_r$ are all on distinct paths from $P_1, ..., P_m$, we contradict the result of step 1. Thus we may assume that for some $j \neq j'$ with $1 \leq j, j' \leq r$, $u_j$ and $u_{j'}$ are both on $P_1$ say, and by symmetry we may assume that either $u_j = u_{j'}$ or $u_j$ occurs before $u_{j'}$ on $P_1$. Consider the sequence

$$u_{j'+1}, v_{j'+1}, ..., u_{j-1}, v_{j-1}, u_j, v_{j'}$$

reading the subscripts modulo $r$ if $j < j'$. The sequence has $2(j - j')$ terms if $j > j'$, and $2r - 2(j' - j)$ terms if $j < j'$. In either case it has fewer than $2r$ terms, and hence contradicts the minimality of $r$.

*Step* 3. *For $u, v \in V(G)$, we say $u < v$ if there is a sequence $u_0, v_0, ..., u_r, v_r$ with $r \geq 0$, such that $u = u_0$, $v_r = v$, and for $1 \leq i \leq r$, $v_{i-1}$, $u_i$ are adjacent, joined by an edge not in any of $P_1, ..., P_m$, and for $1 \leq i \leq r - 1$, $v_i$ occurs after $u_i$ on one of $P_1, ..., P_m$, and $v_0$ occurs after $u_0$ on one of $P_1, ..., P_m$, and either $v_r = u_r$ or $v_r$ occurs after $u_r$ on one of $P_1, ..., P_m$. Then $(V(G), <)$ is a strict partial ordering.*

We must check that if $u < v < w$, then $u < w$; and that $v \not< v$. First, if $u < v < w$, let

$$u = u_0, v_0, ..., u_r, v_r = v,$$
$$v = u'_0, v'_0, ..., u'_s, v'_s = w,$$

be the corresponding sequences. Then the sequence

$$u = u_0, v_0, ..., u_r, v'_0, u'_1, ..., u'_s, v'_s = w$$

demonstrates that $u < w$. Second, if $v < v$, then there is a sequence

$$v = u_0, v_0, ..., u_r, v_r = v;$$

we have $r \geq 1$, since $u_0 \neq v_0$, and so the sequence $u_1, v_1, u_2, v_2, ..., u_{r-1}$, $v_{r-1}, u_r, v_0$ contradicts the result of step 2. Hence $<$ is a strict partial ordering.

*Step* 4. *For each $v \in V(G)$, let $\mu(v)$ be an integer, chosen so that for $v$, $v' \in V(G)$, if $v < v'$, then $\mu(v) < \mu(v')$. (This is possible since $<$ is a strict partial ordering of $V(G)$.) Then $\mu$ satisfies our requirements.*

Certainly if $v'$ occurs on some $P_i$ after $v$, then $v < v'$ and so $\mu(v) < \mu(v')$; hence condition (i) of (2.5) is satisfied. For any integer $t$, let $Z_t = \{v \in V(G)$: for some $i$ $(1 \leq i \leq m)$, $v$ is the first vertex of $P_i$ with $\mu(v) \geq t\}$.

Let $X_t = \{v \in V(G) : \mu(v) < t\}$, and $Y_t = V(G) - (X_t \cup Z_t)$. We must show that $Z_t$ separates $X_t$ and $Y_t$.

But $X_t \cup Y_t \cup Z_t = V(G)$, and so it is sufficient to prove no edge has one [p. 46] end in $X_t$ and the other in $Y_t$. Suppose then that $u \in X_t$ and $v \in Y_t$ are adjacent. Let $v$ be a vertex of $P_1$ say. Since $\mu(v) \geq t$, there is a first vertex $u'$ say of $P_1$ with $\mu(u') \geq t$. We have $u' \in Z_t$ and so $v \neq u'$, and $u'$ occurs on $P_1$ before $v$. But $u$ and $v$ are adjacent, and are joined by an edge not in any of $P_1, ..., P_m$; and so the sequence

$$u', v, u, u$$

demostrates that $u' < u$. Yet $\mu(u') \geq t > \mu(u)$, a contradiction. This completes the proof of (2.5).

(2.6). *Suppose that $G \in \mathscr{F}_\theta$, and $A_1, A_2 \subseteq V(G)$, and $m > 0 \geq \frac{1}{2}\theta^2$ is an integer,*[F6] *and the following conditions are satisfied*:

(i) *there is a unique set $\{P_1, ..., P_m\}$ of $m$ disjoint paths in $G$ from $A_1$ to $A_2$;*

(ii) *every vertex of $G$ is in one of $P_1, ..., P_m$;*

(iii) *$B_1, ..., B_n$ are disjoint connected subgraphs of $G$, and each $B_j$ meets at least $\frac{1}{2}\theta^2$ of $P_1, ..., P_m$.*

*Then $n < 2m\theta^{2\theta-2}\binom{m}{\frac{1}{2}\theta^2}$.*

*Proof.* Define $s = \lceil n/2m \rceil$. Choose an integer-valued function $\mu$ on $V(G)$ as in (2.5). Define a sequence of integers $k_1, ..., k_s$ as follows:

$$k_1 = \min(k : \text{for some } j \ (1 \leq j \leq n), \mu(v) < k \text{ for all } v \in V(B_j)),$$
$$k_l = \min(k : \text{for some } j \ (1 \leq j \leq n), k_{l-1} \leq \mu(v) < k \text{ for all } v \in V(B_j)) \quad (2 \leq l \leq s).$$

We must show that this is well defined. Let us suppose inductively that $k_{l'}$ is well defined for $1 \leq l' < l$, where $1 \leq l \leq s$, and we shall show that $k_l$ is well defined. If $l = 1$ this is clear, and we assume $l > 1$. For each integer $t$, let $Z_t = \{v \in V(G) :$ for some $i$, $v$ is the first vertex of $P_i$ with $\mu(v) \geq t\}$. For $1 \leq l' < l$, let

$$J_{l'} = \{j : 1 \leq j \leq n, \text{and for some } v \in V(B_j), \mu(v) < k_{l'}\}.$$

[F6] **Degenerate case.** With $m > 0$ only, the statement fails when $n = 0$. For $\theta = 6$, the one-vertex graph, $A_1 = A_2 = V(G)$ and $m = 1$, every hypothesis holds, but the conclusion is $0 < 0$, because $\binom{1}{18} = 0$. The uses of (2.6) satisfy $m \geq \frac{1}{2}\theta^2$: in (2.7), $m = \frac{1}{2}\theta^2$, and in (2.8), $m = \frac{1}{2}s\theta^2$ with $s \geq 1$ by the hypothesis added to (2.8).

Then clearly for $l' \geq 2$, we have $J_{l'-1} \subset J_{l'}$; we claim that

$$|J_{l'} - J_{l'-1}| \leq 2m.$$

To prove that it is sufficient to show that for every $j \in J_{l'} - J_{l'-1}$, $B_j$ meets $(Z_{k_{l'}-1} \cup Z_{k_{l'}})$, because $|Z_t| \leq m$ for every integer $t$, and $B_1, ..., B_n$ are disjoint. Suppose then that $j \in J_{l'} - J_{l'-1}$. There exists $u \in V(B_j)$ such that $\mu(u) < k_{l'}$. If there exists $u' \in V(B_j)$ such that $\mu(u') \geq k_{l'}$, then $B_j$ uses a [p. 47] vertex from $Z_{k_{l'}}$, because $Z_{k_{l'}}$ separates $\{v\colon \mu(v) < k_{l'}\}$ and $\{v\colon \mu(v) \geq k_{l'}\}$. We may assume therefore that $k_{l'-1} \leq \mu(v) < k_{l'}$ for every vertex $v$ of $B_j$. By the minimality of $k_{l'}$, there is a vertex $v$ of $B_j$ with $\mu(v) \geq k_{l'} - 1$; and so $\mu(v) = k_{l'} - 1$. But then $v \in Z_{k_{l'}-1}$, and again our claim is true. Thus for $2 \leq l' < l$ we have $|J_{l'} - J_{l'-1}| \leq 2m$. A similar argument shows that $|J_1| \leq 2m$. It follows that

$$|J_{l-1}| \leq 2(l-1)m$$

and hence that $J_{l-1} \neq \{1, ..., n\}$, since $l \leq s = \lceil n/2m \rceil$. Thus $k_l$ is well defined. Hence by induction, $k_1, ..., k_s$ are well defined.

Choose $j_1$ with $1 \leq j_1 \leq n$ so that $\mu(v) < k_1$ for every vertex $v$ of $B_{j_1}$; and for $2 \leq l \leq s$, choose $j_l$ with $1 \leq j_l \leq n$ so that $k_{l-1} \leq \mu(v) < k_l$ for every vertex $v$ of $B_{j_l}$. Put $C_l = B_{j_l}$ $(1 \leq l \leq s)$. Then $C_1, ..., C_s$ are disjoint connected subgraphs of $G$, and each meets at least $\frac{1}{2}\theta^2$ of $P_1, ..., P_m$: and moreover, for $1 \leq l < l' \leq s$, all vertices of $C_l$ on $P_i$ occur before all vertices of $C_{l'}$. By (2.4), $s < \theta^{2\theta-2}\binom{m}{\frac{1}{2}\theta^2}$, and so $n < 2m\theta^{2\theta-2}\binom{m}{\frac{1}{2}\theta^2}$, as required.

(2.7). *Suppose that there are $m$ disjoint paths in $G \in \mathscr{F}_\theta$ from $A_1$ to $A_2$, where $A_1, A_2 \subseteq V(G)$; and that $B_1, ..., B_n$ are disjoint connected subgraphs of $G$, and for $1 \leq j \leq n$, $V(B_j)$ separates $A_1$ and $A_2$. Then either $m < \frac{1}{2}\theta^2$ or $n < \theta^{2\theta}$.*

*Proof.* If possible, choose a counterexample with $|V(G)| + |E(G)| + m$ minimum. Then evidently we have $m = \frac{1}{2}\theta^2$.

Let $\{P_1, ..., P_m\}$ be a set of $m$ disjoint paths in $G$ from $A_1$ to $A_2$. If some $P_i$ and some $B_j$ have an edge in common, we can contract that edge and produce a smaller counterexample. Thus no edge belongs to both some $P_i$ and some $B_j$. Moreover, every edge belongs either to some $P_i$ or to some $B_j$, for otherwise we could delete it.

It follows that

$$E(P_1) \cup \cdots \cup E(P_m) = E(G) - (E(B_1) \cup \cdots \cup E(B_n)).$$

If $\{P'_1, ..., P'_m\}$ is another set of $m$ disjoint paths from $A_1$ to $A_2$, we have, by the same argument,

$$E(P'_1) \cup \cdots \cup E(P'_m) = E(G) - (E(B_1) \cup \cdots \cup E(B_n))$$

and hence

$$E(P'_1) \cup \cdots \cup E(P'_m) = E(P_1) \cup \cdots \cup E(P_m).$$

It follows that $\{P'_1, ..., P'_m\} = \{P_1, ..., P_m\}$, and so the set $\{P_1, ..., P_m\}$ is unique.

We claim that every vertex $v$ of $G$ is in some $P_i$. For if not, and $v$ is an [p. 48] isolated vertex, we can delete $v$, and if $v$ is not an isolated vertex, we can contract some edge incident with ~~$v_0$~~$v$;[F7] in either case we produce a smaller counterexample, which is impossible.

For $1 \le j \le n$, each $P_i$ meets $B_j$, by our hypothesis. By (2.6),

$$n < 2m\theta^{2\theta-2}\binom{m}{\frac{1}{2}\theta^2} = \theta^{2\theta},$$

a contradiction, as required.

(2.8). *Let $A_1, A_2, X \subseteq V(G)$, where $G \in \mathscr{F}_\theta$, and let $B_1, ..., B_n$ be disjoint subgraphs of $G$, each with at most $d$ components. Suppose that the following conditions are satisfied*:

(i) *$|X| \le rn$, for some integer $r \ge 0$;*

(ii) *for $1 \le j \le n$, $X \cup V(B_j)$ separates $A_1$ and $A_2$;*

(iii) *there are $m$ disjoint paths in $G$ from $A_1$ to $A_2$, each avoiding $X$.*

*Suppose also that $d + r > 0$.*[F8] *Let $s = d + r(\theta^2 - 1)$. Then either $m < \frac{1}{2}s\theta^2$ or $n < 2^{s\theta^2/2}(\theta^2 + 2s\theta^{2\theta})$.*

*Proof.* If possible, choose a counterexample with $|V(G)| + |E(G)| + m$ minimum. Then evidently we have $m = \frac{1}{2}s\theta^2$.

Let $P_1, ..., P_m$ be disjoint paths of $G$ from $A_1$ to $A_2$, avoiding $X$. If some edge of $G$ in both in some $P_i$ and in some $B_j$, we may contract it, and if some edge is neither in some $P_i$ nor in some $B_j$, we may delete it, in either case producing a smaller counterexample. It follows that

$$E(P_1) \cup \cdots \cup E(P_m) = E(G) - (E(B_1) \cup \cdots \cup E(B_n)),$$

[F7] **Misprint.** No $v_0$ is defined here, and the vertex meant is $v$.

[F8] **Degenerate case.** The statement fails when $d = r = 0$, so that $s = 0$. Take $A_1 = A_2 = X = \varnothing$, $m = 0$, $n = \theta^2$ and every $B_j$ empty. Then (i)–(iii) hold, but $m = 0 \not< \frac{1}{2}s\theta^2 = 0$ and $n = \theta^2 \not< 2^0(\theta^2 + 0)$. The statement holds when $d + r > 0$. The proof of (2.9) applies (2.8) with $d = r$, and there $r \ge 1$, because for $r = 0$ alternative (i) of (2.9) holds with $J = \varnothing$.

and so $\{P_1, ..., P_m\}$ is the unique set of $m$ disjoint paths of $G$ from $A_1$ to $A_2$ avoiding $X$.

Every vertex $v$ of $V(G) - X$ is in one of $P_1, ..., P_m$; for if not, and $v$ is isolated, we can delete it, and if $v$ is not isolated, we can contract some edge incident with it, in either case producing a smaller counterexample.

Let $J_1 \subseteq \{1, ..., n\}$ be defined by $j \in J_1$ of and only if some component of $B_j \backslash X$ meets at least $\frac{1}{2}\theta^2$ of the paths $P_1, ..., P_m$. By setting $A_1' = A_1 - X$, $A_2' = A_2 - X$, and applying (2.6) to $G \backslash X$, we deduce that $|J_1| < 2m\,\theta^{2\theta-2}2^m$.

Now let $J_2 \subseteq \{1, ..., n\}$ be defined by $j \in J_2$ if and only if $|X \cap V(B_j)| > 2r$. Since $B_1, ..., B_n$ are disjoint, we have

$$rn \geq |X| \geq \sum_{1 \leq j \leq n} |X \cap V(B_j)| \geq 2r\,|J_2|.$$

If $r = 0$, then $X = \varnothing$ and so $J_2 = \varnothing$ by definition. If $r > 0$, then $n \geq 2\,|J_2|$. Thus in either case $|J_2| \leq \frac{1}{2}n$, and so there exists $J_3 \subseteq \{1, ..., n\}$ with

$$|J_3| \geq \tfrac{1}{2}n - 2m\,\theta^{2\theta-2}2^m$$

[p. 49] such that for every $j \in J_3$, $|X \cap V(B_j)| \leq 2r$ and every component of $B_j \backslash X$ meets fewer than $\frac{1}{2}\theta^2$ different $P_i$'s.

Let $j \in J_3$. Let $X \cap V(B_j) = Y_j$, and let $N(Y_j)$ be the set of those vertices which are adjacent in $G$ to at least one element of $Y_j$. A component of $B_j \backslash X$ which includes no element of $N(Y_j)$ is a component of $B_j$, and hence $B_j \backslash X$ has at most $d$ such components. Each of these components meets fewer than $\frac{1}{2}\theta^2$ paths $P_i$. Hence at least $m - \frac{1}{2}\,d\theta^2 \geq \frac{1}{2}r\theta^2(\theta^2 - 1)$ paths $P_i$ meet at least one component of $B_j \backslash X$ which includes an element of $N(Y_j)$. Since $|Y_j| \leq 2r$, there exists $v_j \in Y_j$ such that at least $\frac{1}{4}\theta^2(\theta^2 - 1)$ paths $P_i$ meet a component of $B_j \backslash X$ which includes a vertex adjacent to $v_j$. Let $\{D_j^1, ..., D_j^{\alpha(j)}\}$ be a minimal set of components of $B_j \backslash X$ such that each $D_j^k$ includes a vertex adjacent to $v_j$ and at least $\frac{1}{4}\theta^2(\theta^2 - 1)$ paths $P_i$ meet $D_j^1 \cup \cdots \cup D_j^{\alpha(j)}$. Then $\alpha(j) \geq \frac{1}{2}\theta^2$ since each component of $B_j \backslash X$ meets fewer than $\frac{1}{2}\theta^2$ paths $P_i$. By the minimality of $\{D_j^1, ..., D_j^{\alpha(j)}\}$ there exist $P_{i(1)}, ..., P_{i(\alpha(j))}$ such that $P_{i(u)}$ meets $D_j^u$ but meets no $D_j^t$ $(t \neq u)$, for $u = 1, ..., \alpha(j)$. Then $i(1), ..., i(\alpha(j))$ are distinct: define $I(j) = \{i(1), ..., i(\frac{1}{2}\theta^2)\}$.

Then $I(j) \subseteq \{1, ..., m\}$, for all $j \in J_3$, and so there exists $J_4 \subseteq J_3$ with $|J_4| \geq 2^{-m}|J_3|$, such that for $j, j' \in J_4$, $I(j) = I(j') = I$ say. Renumber, for simplicity, so that $I = \{1, 2, ..., \frac{1}{2}\theta^2\}$, and for $j \in J_4$, $P_i$ meets $D_j^i$ but does not meet any $D_j^{i'}$ for distinct $i, i' \in I$. For each $i \in I$, let $Q_i$ be the subgraph of $G$ consisting of the vertices and edges in the graphs $P_i$ and $D_j^i (j \in J_4)$.

Then $Q_1, ..., Q_{\theta^2/2}$ are all connected, and are disjoint, and do not intersect $X$. Moreover, if $1 \le i \le \frac{1}{2}\theta^2$ and $j \in J_4$, $v_j$ is adjacent to a vertex of $Q_i$. It follows that $G$ has a minor isomorphic to the complete bipartite graph

$$K_{|J_4|, \theta^2/2}.$$

But $K_{\theta^2/2, \theta^2/2}$ has a minor isomorphic to the $\theta$-grid, and $G \in \mathscr{F}_\theta$; thus $|J_4| < \frac{1}{2}\theta^2$. Hence $|J_3| < 2^m \frac{1}{2}\theta^2$, and so

$$\tfrac{1}{2}n - 2m\theta^{2\theta-2}2^m < 2^m \tfrac{1}{2}\theta^2,$$

that is,

$$n < 2^m(\theta^2 + 4m\theta^{2\theta-2}).$$

The result follows.

(2.9). *If $B_1, ..., B_n$ are disjoint connected subgraphs of a graph $G \in \mathscr{F}_\theta$, and $V' \subseteq V(G)$ and $r \ge 0$ is an integer, then one of the following is true*:

(i) *there exists $J \subseteq \{1, ..., n\}$ with $|J| = r$ such that for each $j \in J$ there is a path $P_j$ in $G$ from $B_j$ to $V'$, and the paths $P_j$ $(j \in J)$ are disjoint, and each has no internal vertex in $V' \cup \bigcup_{j \in J} V(B_j)$;*

[p. 50] (ii) *there exist $X \subseteq V(G)$ and $J \subseteq \{1, ..., n\}$ with $|X| + |J| \le 2^{r\theta^4}$, such that $X$ separates $V'$ from every $B_j$ $(j \in \{1, ..., n\} - J)$.*

*Proof.* We assume (i) is false. Define $\nu = 2^{r\theta^4/2}(\theta^2 + 2r\theta^{2\theta+2})$. Let $X_0 = J_0 = \varnothing$. We define inductively a sequence $X_0, X_1, ..., X_\nu$ of subsets of $V(G)$ and a sequence $J_0, J_1, ..., J_\nu$ of subsets of $\{1, ..., n\}$, as follows: Suppose that $X_0, ..., X_{i-1}, J_0, ..., J_{i-1}$ are defined. Put $J' = \{1, ..., n\} - (J_0 \cup \cdots \cup J_{i-1})$. Now no $J \subseteq J'$ satisfies (i), and so (by a form of Menger's theorem) there exist $X_i \subseteq V(G)$ and $J_i \subseteq J'$ with $|X_i| + |J_i| < r$, such that every path in $G$ from any $B_j$ $(j \in J')$ to $V'$ uses either some vertex of $X_i$ or some vertex of $\bigcup_{j \in J_i} V(B_j)$. This completes our inductive definition of $X_i, J_i$.

Put $X' = X_1 \cup \cdots \cup X_\nu$, $J = J_1 \cup \cdots \cup J_\nu$. For $1 \le i \le \nu$, let $B'_i$ be the union of the graphs $B_j$ $(j \in J_i)$. Then $B'_i$ has at most $r$ components. Put

$$A_1 = \bigcup (V(B_j) \colon j \in \{1, ..., n\} - J)$$

and $A_2 = V'$. Now for $1 \le i \le \nu$, $X' \cup V(B'_i)$ separates $A_1$ and $A_2$; for every path from $A_1$ to $A_2$ uses either a vertex of $X_i$ or a vertex of $B'_i$, by definition of $X_i, J_i$. Moreover, for $1 \le i \le \nu$, $|X_i| + |J_i| < r$, and so $|X'| < r\nu$. By (2.8), the maximum number of disjoint paths in $G$ from $A_1$ to $A_2$, avoiding

$X'$, is less than $\frac{1}{2}r\theta^4$ (to apply (2.8), we set $d = r$). By Menger's theorem, there exists $X'' \subseteq V(G)$ with

$$|X''| \leq \tfrac{1}{2}r\theta^4$$

such that $X' \cup X''$ separates $A_1$ and $A_2$. Put $X = X' \cup X''$; then $X$ separates $V'$ from every $B_j$ ($j \in \{1, ..., n\} - J$); and

$$\begin{aligned} |X| + |J| &\leq |X''| + |X'| + |J| \\ &\leq |X''| + \sum_{1 \leq i \leq \nu} (|X_i| + |J_i|) \\ &\leq \tfrac{1}{2}r\theta^4 + r\nu \\ &\leq 2^{r\theta^4} \end{aligned}$$

(after some arithmetic) and so (ii) is true.

Incidentally, the following extension of (2.7) will appear in [4].

(2.10). *There is an integer $\theta' > 0$ such that if $A_1, ..., A_{\theta'}$ are disjoint connected subgraphs of a graph $G \in \mathscr{F}_\theta$, and $B_1, ..., B_{\theta'}$ are also disjoint connected subgraphs of $G$, then some $A_i$ is disjoint from some $B_j$.*

## [p. 51] 3. The Main Theorem

In order to prove our main theorem (1.1), it is only necessary to prove a special case, as we shall now see. Let $\mathbf{Y}_1$ be the complete bipartite graph $K_{1,3}$. For $\lambda \geq 2$, we define $\mathbf{Y}_\lambda$ inductively by taking a copy of $\mathbf{Y}_{\lambda-1}$, and to each vertex of valency 1 in this graph making adjacent two new vertices. (See Fig. 1.)

In order to prove (1.1), it is only necessary to prove the following:

(3.1). *For $\lambda \geq 1$, odd, there is a number $w(\lambda)$ such that every graph with no minor isomorphic to $\mathbf{Y}_\lambda$ has path-with $\leq w(\lambda)$.*

(The "odd" condition is introduced for future technical convenience.)

*Proof of* (1.1) (assuming (3.1)). For any forest $H$ there is an odd value of $\lambda$ such that $\mathbf{Y}_\lambda$ has a minor isomorphic to $H$ (e.g., any odd $\lambda \geq |V(H)|$ will do, although this is usually extravagent). If $G$ is a graph with no minor isomorphic to $H$, then it certainly has no minor isomorphic to $\mathbf{Y}_\lambda$, and so, by (3.1), it has path-width $\leq w(\lambda)$. Thus (1.1) is true.

We now introduce a second class of graphs. $\mathbf{H}_0$ is the graph with just one vertex and no edges, and $\mathbf{H}_1$ is $K_{1,2}$. For $\lambda \geq 2$, we define $\mathbf{H}_\lambda$ inductively by

taking a copy of $\mathbf{H}_{\lambda-1}$, and to each vertex of valency 1 in this graph making adjacent two new vertices. (See Fig. 2.)

(3.2). *For any even $\lambda \geq 0$, the $(2^{(\lambda+1)/2}\lambda/2+1 - 1)$-grid has a minor isomorphic to $\mathbf{H}_\lambda$.*[F9]

~~The proof by induction is left to the reader.~~ We hope that a "proof by diagram" is convincing. Figure 3 shows a subdivision of $\mathbf{H}_6$ drawn as a subgraph of a 15-grid.

We give the induction. For $n \geq 0$ let $s_n = 2^{n+1} - 1$ and $m_n = 2^n$. We call $v_{m_n m_n}$ the *middle* of the $s_n$-grid, and $\{v_{im_n} : 1 \leq i \leq s_n\}$ its *middle column*. The *root* of $\mathbf{H}_{2n}$ is its vertex of even valency (its only vertex if $n = 0$). We prove by induction on $n$ that there are sets $Y(v)$ $(v \in V(\mathbf{H}_{2n}))$ of vertices of the $s_n$-grid satisfying (M1)–(M3) below (with $G_1 = \mathbf{H}_{2n}$ and $G_2$ the grid), such that $Y(v)$ is the set containing only the middle when $v$ is the root, and no other $Y(v)$ meets the middle column. Such sets give a minor of the grid isomorphic to $\mathbf{H}_{2n}$, and this proves (3.2). For $n = 0$ the grid has one vertex, and we take $Y$ of the root to be that vertex.

Now let $n \geq 0$, put $s = s_n$ and $m = m_n$, and assume the result for $n$. Then $s_{n+1} = 2s+1$, and the middle of the $(2s+1)$-grid is $v_{s+1,s+1}$. Deleting row $s+1$ and column $s+1$ from the $(2s+1)$-grid leaves four disjoint $s$-grids, its *quarters*, with middles $v_{mm}$, $v_{s+1+m,m}$, $v_{m,s+1+m}$ and $v_{s+1+m,s+1+m}$. The tree $\mathbf{H}_{2n+2}$ consists of its root $\rho$, two neighbours $a, b$ of $\rho$, and four copies of $\mathbf{H}_{2n}$, where $a$ is adjacent to the roots of two of the copies and $b$ to the roots of the other two. Put each copy in one quarter, by the sets given by the inductive hypothesis, with the copies at $a$ in the two quarters with $j \leq s$. Let

$$A = \{v_{im} : m+1 \leq i \leq s+m\} \cup \{v_{s+1,j} : m \leq j \leq s\},$$
$$B = \{v_{i,s+1+m} : m+1 \leq i \leq s+m\} \cup \{v_{s+1,j} : s+2 \leq j \leq s+1+m\},$$

and let $Y(\rho) = \{v_{s+1,s+1}\}$, $Y(a) = A$, $Y(b) = B$. The sets $A$ and $B$ are disjoint, and each induces a connected subgraph, being the union of a vertical and a horizontal path which share a vertex in row $s+1$. They do not meet column $s+1$. They meet the quarters only in vertices of the middle columns of the quarters other than their middles, and these vertices are in no set of the four copies, by the inductive hypothesis. The set $A$ contains neighbours of $v_{s+1,s+1}$, of $v_{mm}$ and of $v_{s+1+m,m}$, namely $v_{s+1,s}$, $v_{m+1,m}$ and $v_{s+m,m}$.

[F9] **Misprint.** For even $\lambda$ the printed exponent $(\lambda+1)/2$ is not an integer. The size meant is $2^{\lambda/2+1} - 1$: it is 1 for $\lambda = 0$ and 15 for $\lambda = 6$, as in Figure 3. Putting $\lambda + 1$ for $\lambda$ gives the exponent $(\lambda+3)/2$ of (3.4), which is printed correctly.

Similarly $B$ contains neighbours of $v_{s+1,s+1}$ and of the middles of the two other quarters. Hence the sets satisfy (M1)–(M3) for $\mathbf{H}_{2n+2}$. Moreover $Y(\rho)$ is the middle of the $(2s+1)$-grid, and no other set meets column $s+1$, which is its middle column. This completes the induction. [F10]

(3.3). *For any $\lambda \geq 1$, $\mathbf{Y}_\lambda$ is isomorphic to a minor of $\mathbf{H}_{\lambda+1}$.*

The proof is clear. To see this, let $\rho$ be the root of $\mathbf{H}_{\lambda+1}$ (its vertex of valency 2). Deleting $\rho$ and its two neighbours from $\mathbf{H}_{\lambda+1}$ leaves four components, each isomorphic to $\mathbf{H}_{\lambda-1}$, and each containing one vertex at distance 2 from $\rho$, which is its root. Contract the two edges incident with $\rho$, and delete one of the four components. In the resulting tree the new vertex is adjacent to the roots of the three remaining components, and so the tree is isomorphic to $\mathbf{Y}_\lambda$, which consists of a vertex adjacent to the roots of three copies of $\mathbf{H}_{\lambda-1}$.[F11]

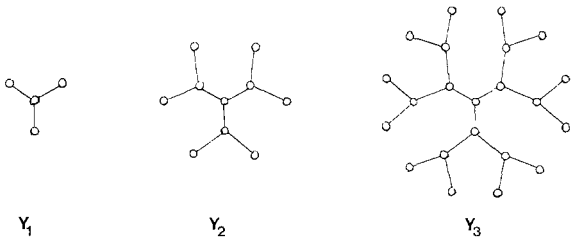
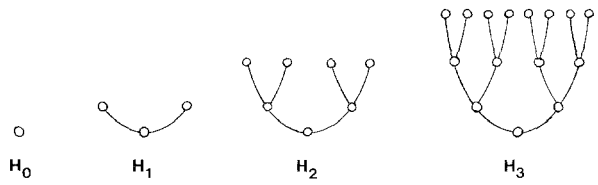



FIGURE 1




[p. 52]

FIGURE 2

(3.4). *For any odd $\lambda \geq 1$, the $(2^{(\lambda+3)/2}-1)$-grid has a minor isomorphic to $\mathbf{Y}_\lambda$.*

This follows from (3.2) and (3.3).

[F10] **Missing proof.** The paper leaves the induction to the reader and shows the case $\lambda = 6$ in Figure 3. The induction above is the construction of Figure 3, and it is the one used in the formalization.

[F11] **Missing proof.** The paper says that the proof is clear. The argument is given because (3.3) is also used for the observation (b) after (1.2). It is the one used in the formalization.

(3.5). *For odd $\lambda \geq 1$, let $\theta = \max(6, 2^{(\lambda+3)/2})$. For any graph $G$, if $G$ has no minor isomorphic to $\mathbf{Y}_\lambda$, then $G \in \mathscr{F}_\theta$.*

This follows from (3.4).

Thus to study the graphs with no minor isomorphic to $\mathbf{Y}_\lambda$, we can confine ourselves to $\mathscr{F}_\theta$, and hence we can apply the results of Section 2.

A *rooted* graph is a graph with one vertex distinguished, called the *root*. We denote the root of $G$ by $\rho(G)$.

If $G_1, G_2$ are rooted graphs we say that $G_2$ has a *rooted minor isomorphic to* $G_1$ if for each vertex $v$ of $G_1$ there is a subset $Y(v)$ of the vertices of $G_2$, satisfying conditions (M1)–(M4):

(M1) *For distinct $v, v' \in V(G_1)$, $Y(v) \cap Y(v') = \varnothing$.*

(M2) *For each $v \in V(G_1)$, $G_2 \,|\, Y(v)$ is connected.*

(M3) *There is an injection $f\colon E(G_1) \to E(G_2)$ such that for every edge*

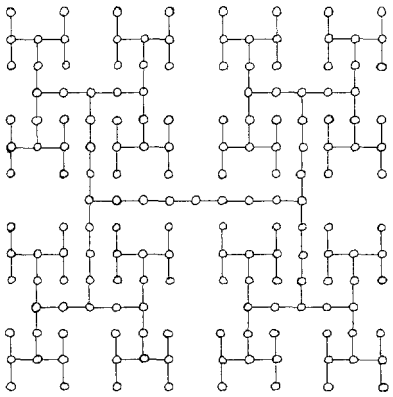


FIGURE 3

[p. 53] *$e$ of $G_1$, if $e$ has ends $v, v'$ say, then $f(e)$ has one end in $Y(v)$ and the other in $Y(v')$.*

(M4) $\rho(G_2) \in Y(\rho(G_1))$.

We now introduced two further classes of trees. If $\gamma \geq 1$, $\delta \geq 1$ are integers, the rooted tree $P(\gamma, \delta)$ is defined as follows: Take $\delta$ disjoint copies of $\mathbf{H}_{\gamma-1}$. Each of these copies has a unique vertex of even valency; let these be $u_1, ..., u_\delta$, say. Take a path of $\delta$ new vertices $v_1, ..., v_\delta$ say, and for $1 \leq i \leq \delta$, add an edge joining $u_i$ and $v_i$. The resultant graph is rooted at $v_1$, and is $P(\gamma, \delta)$. We also define $P(0, \delta)$ to be the path of $\delta$ vertices, rooted at one end, for $\delta \geq 1$.

Second, if $\gamma \geq 1$, $\delta \geq 2$ are integers, we define a rooted tree $Q(\gamma, \delta)$ as follows: Take a copy of $P(\gamma-1, \delta-1)$, with root $v_1$ say, and a disjoint copy of $\mathbf{H}_{\gamma-1}$. Let $v_2$ be the (unique) vertex of the second graph with even valency. Take a new vertex $v$, and two new edges, joining $v$ to $v_1, v_2$, respectively. The resultant graph is $Q(\gamma, \delta)$, and it is rooted at $v$. (See Fig. 4.)

Now let $\lambda \geq 1$ be odd. Define $\theta = \max(6, 2^{(\lambda+3)/2})$, $\alpha = 2^{\theta^4}$, and $\beta = 2^{\theta^6}$. For integers $x, i \geq 0$, define $f_i(x)$ inductively, by

$$f_0(x) = x, \qquad f_i(x) = f_{i-1}(\beta^x)(i > 0).$$

For integers $\gamma, \delta$ with $\gamma \geq 0$, and $\delta \geq 2$, define $p(\gamma, \delta) = f_\gamma(\delta)$; and if $\gamma \geq 1$, define $q(\gamma, \delta) = f_{\gamma-1}(\beta + 2\delta)$.

(3.6). *With the above definitions, the following inequalities hold*:

(i) $p(0, \delta) \geq \delta - 2$ *for all* $\delta \geq 2$;

(ii) $q(\gamma, \delta) \geq \theta^2 + p(\gamma - 1, 2 + 4\theta^{2\theta} + 2\delta)$ *for* $1 \leq \gamma$, $2 \leq \delta$;

(iii) $p(\gamma, \delta) \geq q(\gamma, 4)$ *for* $1 \leq \gamma$, $2 \leq \delta$;

(iv) $p(\gamma, \delta) \geq \alpha^{\delta \cdot 2^\gamma} + \max(p(\gamma, \delta - 1), q(\gamma, 3\alpha^{\delta \cdot 2^\gamma} + 6))$ *for* $1 \leq \gamma \leq \lambda$, $3 \leq \delta$.

*Proof.* This is routine~~, and is left to the reader~~. By definition, $f_i(x) = f_{i-1}(\beta^x)$ for $i \geq 1$ and $x \geq 0$.

(1) *For each integer* $i \geq 0$, $f_i$ *is strictly increasing, and* $f_i(x + d) \geq f_i(x) + d$ *for all integers* $x, d \geq 0$.

The map $f_0$ is the identity. For $i > 0$, $f_i$ is the map $x \mapsto \beta^x$ followed by $f_{i-1}$. The first is strictly increasing since $\beta = 2^{\theta^6} > 1$, and so $f_i$ is strictly increasing, by induction on $i$. A strictly increasing map from the non-negative integers to themselves increases by at least 1 from $x$ to $x + 1$, and so by at least $d$ from $x$ to $x + d$. This proves (1).

(2) $\beta \geq 16$, *and* $\beta \geq \theta^2 + 2 + 4\theta^{2\theta}$.

Since $\theta \geq 6$, we have $\beta = 2^{\theta^6} \geq 16$ and $\theta^6 = \theta^2\theta^4 \geq 3\theta^2 \geq 2\theta^2 + 3$. Since $\theta \leq 2^\theta$, we have $\theta^{2\theta} \leq 2^{2\theta^2}$. Since also $\theta^2 \leq \theta^{2\theta}$, it follows that $\theta^2 + 2 + 4\theta^{2\theta} \leq 8\theta^{2\theta} \leq 2^{2\theta^2+3} \leq 2^{\theta^6} = \beta$. This proves (2).

Now (i) holds since $p(0, \delta) = \delta$. For (ii), let $y = 2 + 4\theta^{2\theta} + 2\delta$. By (2), $\beta + 2\delta \geq y + \theta^2$, and so by (1), $q(\gamma, \delta) = f_{\gamma-1}(\beta + 2\delta) \geq f_{\gamma-1}(y) + \theta^2 = p(\gamma - 1, y) + \theta^2$. For (iii), $\beta^\delta \geq \beta^2 \geq \beta + 8$ by (2), and so $p(\gamma, \delta) = f_{\gamma-1}(\beta^\delta) \geq f_{\gamma-1}(\beta + 8) = q(\gamma, 4)$. For (iv), let $A = \alpha^{\delta \cdot 2^\gamma}$. Since $\gamma \leq \lambda$ and $\theta \geq 2^{(\lambda+3)/2}$, we have $\theta^2 \geq 2^{\lambda+3} \geq 2^{\gamma+1}$. Since also $\delta \geq 2$, it follows that $\delta \cdot 2^\gamma \theta^4 \leq \frac{1}{2}\delta\theta^6 \leq (\delta-1)\theta^6$. Hence $A \leq 2^{(\delta-1)\theta^6} = \beta^{\delta-1}$. Also $\beta + 12 \leq 2\beta^{\delta-1}$, and so by (2),

$$\beta^\delta \geq 16\beta^{\delta-1} \geq \max(\beta^{\delta-1} + A,\ 7A + \beta + 12).$$

By (1) it follows that $p(\gamma,\delta) = f_{\gamma-1}(\beta^\delta) \geq f_{\gamma-1}(\beta^{\delta-1}) + A = p(\gamma,\delta-1) + A$, and that $p(\gamma,\delta) \geq f_{\gamma-1}(\beta + 6A + 12) + A = q(\gamma, 3A+6) + A$. So $p(\gamma,\delta)$ is at least $A$ plus the larger of $p(\gamma,\delta-1)$ and $q(\gamma,3A+6)$, and this proves (iv).[F12]

The aim of the next two sections is to prove the following:

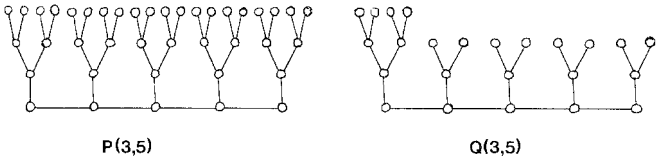



FIGURE 4

[p. 54] (3.7). (i) *If $G \in \mathscr{F}_\theta$, and is rooted and connected, and has no rooted minor isomorphic to $P(\gamma,\delta)$, then $G$ has path-width at most $p(\gamma,\delta)$, for $0 \leq \gamma \leq \lambda$, $\delta \geq 2$.*

(ii) *If $G \in \mathscr{F}_\theta$, and is rooted and connected, and has no rooted minor isomorphic to $Q(\gamma,\delta)$, then $G$ has path-width at most $q(\gamma,\delta)$, for $1 \leq \gamma \leq \lambda$, $\delta \geq 2$.*

*Proof of* (3.1) (assuming (3.7)). Let $G$ be a graph with no minor isomorphic to $\mathbf{Y}_\lambda$, where $\lambda \geq 1$ and is odd.

Let $C$ be any component of $G$, and let $C$ be assigned a root, arbitrarily. Now $C$ has no rooted minor isomorphic to $P(\lambda,3)$, because $P(\lambda,3)$ has a minor isomorphic to $\mathbf{Y}_\lambda$. Moreover, $C \in \mathscr{F}_\theta$, by (3.5). Hence, by (3.7)(i), $C$ has path-width $\leq p(\lambda,3)$, and so by (1.4), $G$ has path-width $\leq p(\lambda,3)$.

Thus, to prove our main theorem, it suffices to prove (3.7).

The proof of (3.7) is divided into three parts, as follows: In Part 1 we prove that (3.7)(i) is true when $\gamma = 0$. In Part 2 we show that for $\gamma$ with $1 \leq \gamma \leq \lambda$, if (3.7)(i) is true for $\gamma - 1$ and all $\delta \geq 2$, then (3.7)(ii) is true for $\gamma$ and all $\delta \geq 2$. Finally, in Part 3 we show that for any $\gamma$ with $1 \leq \gamma \leq \lambda$, if (3.7)(ii) is true for $\gamma$ and all $\delta \geq 2$, then so is (3.7)(i). These three parts combined yield that (3.7) is true.

*Part* 1 ($\gamma = 0$). $P(0,\delta)$ is a $\delta$-vertex path, rooted at one end. We show, by induction on $\delta$, that for $\delta \geq 2$ every connected rooted graph with no rooted minor isomorphic to $P(0,\delta)$ has path-width $\leq \delta - 2$. If $\delta = 2$ this is clear. We suppose then that $\delta > 2$. Let $G_1, ..., G_r$ be the components of $G \backslash \{\rho(G)\}$. For $1 \leq i \leq r$, let $v_i$ be a vertex of $G_i$ adjacent to $\rho(G)$ in $G$.

[F12] **Missing proof.** The paper leaves this proof to the reader. The argument above is the one used in the formalization.

We define $\rho(G_i) = v_i$ $(1 \le i \le r)$. Then certainly for $1 \le i \le r$, $G_i$ has no rooted minor isomorphic to $P(0, \delta - 1)$, and so by induction has path-width $\le \delta - 3$. Hence, by (1.4), $G \backslash \{\rho(G)\}$ has path-width $\le \delta - 3$, and so by (1.5), $G$ has path-width $\le \delta - 2$. This completes the inductive argument that for $\delta \ge 2$ every rooted connected graph with no rooted minor isomorphic to $P(0, \delta)$ has path-width $\le \delta - 2$. But by (3.6)(i), $p(0, \delta) \ge \delta - 2$, and so (3.7)(i) is true if $\gamma = 0$. This completes the argument for Part 1.

## 4. Part 2: The Reduction of $Q(\gamma, \delta)$

We now assume that $1 \le \gamma \le \lambda$, and that for $\delta \ge 2$, every rooted connected graph in $\mathscr{F}_\theta$ with no rooted minor isomorphic to $P(\gamma - 1, \delta)$ has path-width $\le p(\gamma - 1, \delta)$. We shall prove that for all $\delta \ge 2$, every rooted connected graph in $\mathscr{F}_\theta$ with no rooted minor isomorphic to $Q(\gamma, \delta)$ has path-width $\le q(\gamma, \delta)$.

Suppose then that $G \in \mathscr{F}_\theta$, and is rooted and connected, and has no [p. 55] rooted minor isomorphic to $Q(\gamma, \delta)$. Define $\nu = \theta^{2\theta}$, and $\mu = 2 + 4\nu + 2\delta$. If $G$ has no rooted minor isomorphic to $P(\gamma - 1, \mu)$, then by hypothesis it has path-width $\le p(\gamma - 1, \mu) \le q(\gamma, \delta)$, by (3.6)(ii). We assume then that $G$ has a rooted minor isomorphic to $P(\gamma - 1, \mu)$. Choose $N$ maximum such that $G$ has a rooted minor isomorphic to $P(\gamma - 1, N)$, and then $N \ge \mu$. Choose an integer $T$ maximum so that $2\nu T \le N - 2\delta$. Then $T \ge 2$, since $\mu \ge 4\nu + 2\delta$. Put $n - 1 = \nu T$; and then

$$N - 2\delta - 2\nu < 2n - 2 \le N - 2\delta$$

by the maximality of $T$.

Now $G$ has a rooted minor isomorphic to $P(\gamma - 1, N)$; and so, since all vertices of this graph have valency $\le 3$, $G$ has a subgraph $H'$ which is isomorphic to a subdivision of $P(\gamma - 1, N)$, with a vertex $v$ say corresponding to the root of $P(\gamma - 1, N)$, and $G$ has a path from $v$ to $\rho(G)$, with no vertex except $v$ in common with $H'$. Let $H$ be the subgraph of $G$ consisting of the vertices and edges in $H'$ together with those in the path.

Now $P(\gamma - 1, N)$ is formed by taking a path of $N$ vertices $v'_1, ..., v'_N$ say, and if $\gamma \ge 2$, taking $N$ disjoint copies of $\mathbf{H}_{\gamma-2}$, and joining them to $v'_1, ..., v'_N$ appropriately; and then letting $v'_1$ be the root. Let $v_1, ..., v_N$ be the vertices of $H'$ which correspond to $v'_1, ..., v'_N$. For $1 \le j \le N - 1$, let $R_j$ be the set of vertices of $H$ which may be joined to $v_N$ by a path of $H$ which avoids $v_j$, and put $R_0 = V(H)$, $R_N = \varnothing$. For $1 \le j < n$, let $S_j = R_{2j-2} - R_{2j}$; and put $S_n = R_{2n-2}$. Then $S_1, ..., S_n$ are disjoint, and have union $V(H)$. For

$1 \le j \le n$, let $B_j$ be $H \mid S_j$. Then each $B_j$ is a connected subgraph of $G$, and $B_1, ..., B_n$ are disjoint.

(4.1). *For $1 \le j \le n$, $S_j$ separates $\bigcup_{1 \le j' \le j} S_{j'}$ from $\bigcup_{j \le j' \le n} S_{j'}$ in $G$.*

For suppose not; then there is a path of $G$ from $S_{j_1}$ to $S_{j_2}$ say, where $1 \le j_1 < j < j_2 \le n$, with no interior vertices in $H$. Thus there is a path $P$ of $G$ from $S_{j_1}$ to $S_n$, avoiding $S_1 \cup \cdots \cup S_{j_1-1} \cup S_{j_1+1}$. But then $G$ has a rooted minor isomorphic to $Q(\gamma, \delta)$, as can be seen by deleting all vertices and edges of $G$ except those in $H$ and $P$, deleting all the vertices of $S_{j_1+2} \cup \cdots \cup S_{n-1}$ except those in $P$, and contracting all edges in $B_1, ..., B_{j_1}$ and $P$ (note that $B_n$ has a minor isomorphic to $P(\gamma - 1, 2\delta)$ since $N - 2n + 2 \ge 2\delta$).

(4.2). *For $\nu \le j \le n$, there exists $X_j \subseteq V(G)$ with $|X_j| < \frac{1}{2}\theta^2$, separating $\bigcup_{1 \le j' \le j-\nu+1} S_{j'}$ and $\bigcup_{j \le j' \le n} S_{j'}$.*

Suppose not. Then by Menger's theorem there are $\frac{1}{2}\theta^2$ disjoint paths of $G$ from $\bigcup_{1 \le j' \le j-\nu+1} S_{j'}$ to $\bigcup_{j \le j' \le n} S_{j'}$, and yet these two sets are separated [p. 56] by each of $S_{j-\nu+1}, S_{j-\nu+2}, ..., S_j$, by (4.1). Then we have a contradiction from (2.7).

For $\nu \le j \le n$, choose $X_j$ as in (4.2), minimal.

For $1 \le j \le n$, let $Z_j$ denote the set of all vertices of $G$ which can be joined to a vertex of $S_j$ by a path of $G$ which avoids $(S_1 \cup \cdots \cup S_n) - S_j$. Then clearly $Z_j \cap (S_1 \cup \cdots \cup S_n) = S_j$ and $Z_1 \cup \cdots \cup Z_n = V(G)$. By (4.1), $Z_j \cap Z_{j'} \ne \varnothing$ only if $|j - j'| \le 1$.

(4.3). *For $\nu \le j \le n$, $X_j \subseteq S_{j-\nu+1} \cup Z_{j-\nu+2} \cup \cdots \cup Z_{j-1} \cup S_j$.*

For if $v \in X_j$, then by the minimality of $X_j$ there is a path $P$ from $\bigcup_{1 \le j' \le j-\nu+1} S_{j'}$ to $\bigcup_{j \le j' \le n} S_{j'}$ which avoids $X_j - \{v\}$ and uses $v$, and which has no interior vertices in

$$\bigcup_{1 \le j' \le j-\nu+1} S_{j'} \cup \bigcup_{j \le j' \le n} S_{j'}.$$

Since $S_{j-\nu+2} \cup \cdots \cup S_{j-1}$ separates $\bigcup_{1 \le j' \le j-\nu+1} S_{j'}$ and $\bigcup_{j \le j' \le n} S_{j'}$ (by (4.1)), $P$ has an interior vertex $u$ in $S_{j-\nu+2} \cup \cdots \cup S_{j-1}$. Choose $u$ such that the subpath $P'$ of $P$ from $v$ to $u$ has minimum length. Then no vertices of $P'$ except $v$, $u$ are in $S_1 \cup \cdots \cup S_n$. If $v \notin S_1 \cup \cdots \cup S_n$, then $v \in Z_{j-\nu+2} \cup \cdots \cup Z_{j-1}$, because of $P'$. If $v \in S_1 \cup \cdots \cup S_n$, then

$$v \in S_{j-\nu+1} \cup S_{j-\nu+2} \cup \cdots \cup S_{j-1} \cup S_j$$

by (4.1), since no interior vertices of $P'$ are in $S_{j-\nu+1}$ or $S_j$. Thus in either case,

$$v \in S_{j-\nu+1} \cup Z_{j-\nu+2} \cup \cdots \cup Z_{j-1} \cup S_j$$

as claimed.

(4.4). *For $\nu \le i \le n$, $1 \le j \le n$, if $s \in S_j$, then*
(i) *$X_i$ separates $s$ and $\rho(G)$ if $j \ge i$, and*[F13]
(ii) *$X_i$ does not separate $s$ and $\rho(G)$ if $j \le i - \nu$.*

Statement (i) follows from the definition of $X_i$, since $\rho(G) \in S_1$. To show (ii), we observe that from (4.3), the path of $H$ from $s$ to $\rho(G)$ does not meet $X_i$ if $j \le i - \nu$.

Let $Y_i = X_{\nu i}$ for $1 \le i \le T$. From (4.3), $Y_1, ..., Y_T$ are disjoint.

(4.5). *For $1 \le i, i' \le T$ and for every $y \in Y_{i'}$, $Y_i$ separates $y$ and $\rho(G)$ if and only if $i \le i'$.*

[p. 57] If $i = i'$ the result is true, and so we assume that $i \ne i'$. Define

$$Z = S_{\nu i' - \nu + 1} \cup Z_{\nu i' - \nu + 2} \cup \cdots \cup Z_{\nu i' - 1} \cup S_{\nu i'}$$

and

$$S = S_{\nu i' - \nu + 1} \cup S_{\nu i' - \nu + 2} \cup \cdots \cup S_{\nu i' - 1} \cup S_{\nu i'}.$$

By (4.3), $y \in Z$ and there is a path of $G$ from $y$ to some $s \in S$ within $Z$. Now $Z_j \cap Z_{j'} \ne \varnothing$ only if $|j - j'| \le 1$, and so this path avoids

$$S_{\nu i - \nu + 1} \cup Z_{\nu i - \nu + 2} \cup \cdots \cup Z_{\nu i - 1} \cup S_{\nu i}$$

since $i \ne i'$. Hence by (4.3) it avoids $Y_i$. Thus $Y_i$ separates $y$ and $\rho(G)$ if and only if it separates $s$ and $\rho(G)$. The result follows from (4.4).

For $1 \le i \le T - 1$, let $W_i$ be the set of all vertices $v$ of $G$ such that $Y_i$ separates $v$ and $\rho(G)$, and $Y_{i+1}$ does not separate $v$ and $\rho(G)$. Let $W_0$ be the set of vertices not separated from $\rho(G)$ by $Y_1$, and let $W_T$ be the set of vertices separated from $\rho(G)$ by $Y_T$. Then the sequence

$$W_0 \cup Y_1, W_1 \cup Y_2, W_2 \cup Y_3, ..., W_{T-1} \cup Y_T, W_T$$

is a path-decomposition of $G$ by (4.5) and (1.7), and the intersection of any consecutive pair of terms of this sequence has cardinality $\le \frac{1}{2}\theta^2$. In order to complete step 2, it suffices to show that $G \mid W_0, ..., G \mid W_T$ each have path-width $\le p(\gamma - 1, \mu)$, by (1.6) and (3.6)(ii). By (1.4), it suffices to show that for $0 \le i \le T$, every component of $G \mid W_i$ has path-width $\le p(\gamma - 1, \mu)$.

[F13] **Misprint.** As printed, (i) and (ii) contradict each other for $i = n$ and $s = \rho(G) \in S_1$, since $1 \le n - \nu$, and $\rho(G) \notin X_n$ by (4.3). The proof of (i) works for $j \ge i$: $X_i$ separates $S_1 \cup \cdots \cup S_{i-\nu+1}$, which contains $\rho(G)$, from $S_i \cup \cdots \cup S_n$, which contains $s$. The paper uses (i) only for such $j$, in (4.5) and (4.6).

Thus, take $0 \le i \le T$, and let $C$ be a component of $G \setminus W_i$. Assign a root to $C$, arbitrarily. By our initial hypothesis it is sufficient to show that $C$ has no rooted minor isomorphic to $P(\gamma-1,\mu)$. Suppose for a contradiction, that it does; then $C$ has a subgraph $D$ which is isomorphic to a subdivision of $P(\gamma-1,\mu)$.

Let

$$A = \bigcup_{\nu(i+1)\le j\le n} S_j, \quad B = \bigcup_{1\le j\le \nu(i-1)} S_j.$$

(4.6). *$A, B$ and $W_i$ are disjoint.*

For every vertex of $A$ is separated from $\rho(G)$ by $Y_i$ (if $i > 0$) and by $Y_{i+1}$ (if $i < T$), from (4.4). Every vertex of $W_i$ is separated from $\rho(G)$ by $Y_i$ (if $i > 0$) but not by $Y_{i+1}$ (if $i < T$), by the definition of $W_i$. Every vertex of $B$ is separated from $\rho(G)$ neither by $Y_i$ (if $i > 0$) nor by $Y_{i+1}$ (if $i < T$), from (4.4). The result follows.

[p. 58] (4.7). *There is a path in $G$ from $V(D)$ to $V(H)$ avoiding $A \cup B$.*

This follows from (4.5) and the definition of $W_i$ when $i = 0$. When $1 \le i \le T$ there is a path $P$ within $W_i$ from $D$ to $Y_i$, by definition of $W_i$. $P$ certainly avoids $A \cup B$, by (4.6). Let $y$ be a vertex of $Y_i$ on $P$. By (4.3) there is a path $P'$ from $y$ to $S_{\nu i-\nu+1} \cup \cdots \cup S_{\nu i}$ which avoids $A \cup B$. The union of $P$ and $P'$ yields a path satisfying (4.7).

If $i \ge 2$, let $w_1$ be the vertex of $S_{\nu i-\nu}$ which is adjacent in $H$ to a vertex of $S_{\nu i-\nu+1}$. If $i = 0$ or 1, let $w_1$ be $\rho(G)$.

(4.8). *There is a path $P_1$ in $G$ from $D$ to $w_1$, avoiding $A \cup B - \{w_1\}$.*

This is clear from (4.7). Let $P_1$ be a minimal such path, so that it uses only one vertex of $D$, $d_1$ say. Let $P_0$ be the path of $H$ from $w_1$ to $\rho(G)$.

Now $D$ is a subdivision of $P(\gamma-1, 4\nu+2\delta+2)$. But $P(\gamma-1, 4\nu+2\delta+2)$ consists of a $(4\nu+2\delta+2)$-vertex path with vertices $p(1), ..., p(4\nu+2\delta+2)$ say, in order, and (if $\gamma > 1$) $4\nu+2\delta+2$ copies of $\mathbf{H}_{\gamma-2}$, $H(1), ..., H(4\nu+2\delta+2)$ say, in order, and an edge from $p(j)$ to $H(j)$ $(1 \le j \le 4\nu+2\delta+2)$. By the *first end* of $D$ we mean the subgraph of $D$ corresponding to $H(1)$, $H(2)$ and the path of $D$ joining them. By the *second end* we mean the subgraph corresponding to $H(4\nu+2\delta+1)$, $H(4\nu+2\delta+2)$ and the path joining them.

(4.9). *$d_1$ is in one of the ends of $D$.*

Otherwise, since $4\nu+2\delta+2 \ge 2\delta$, $G$ has a rooted minor isomorphic to $Q(\gamma,\delta)$, as can be seen by deleting all vertices and edges of $G$ not in $D, P_0$, or $P_1$, and contracting all edges in $P_0$ and $P_1$.

Without loss of generality we may assume that $d_1$ is in the first end of $D$.

(4.10). *$i \neq T$.*

If $i = T$, then $G$ has a rooted minor isomorphic to $P(\gamma - 1, 2\nu(T-1) + 4\nu + 2\delta)$ by (4.8) and (4.9), as can be seen by deleting all vertices and edges of $G$ except those of $H$ within $B$ and those of $P_1$ and $D$, and by contracting the edges in $P_1$ and those in the first end of $D$. But

$$2\nu(T-1) + 4\nu + 2\delta > N$$

by the maximality of $T$, contrary to the maximality of $N$.

Let $w_2$ be the vertex of $S_{\nu i+\nu}$ adjacent in $H$ to a vertex of $S_{\nu i+\nu-1}$.

(4.11). *There is a path $P_2$ from $D$ to $w_2$ avoiding $A \cup B - \{w_2\}$.*

This follows from (4.7) and (4.10).

Choose a minimal path $P_2$ satisfying (4.11) so that $P_2$ only contains one [p. 59] vertex of $D$, $d_2$ say. Let $P_3$ be the path of $H$ from $w_2$ to $S_n$, with no internal vertex in $S_n$.

(4.12). *$d_2$ is in the second end of $D$.*

If not, then the second end of $D$ is disjoint from $P_1, P_2, A$ and $B$. But then $G$ has a rooted minor isomorphic to $Q(\gamma, \delta)$, as can be seen by deleting all vertices and edges of $G$ not in $D$, $P_0$, $P_1$, $P_2$, $P_3$, and $S_n$ and contracting the edges of $P_0$, $P_1$, $P_2$, $P_3$, and those of $D$ not incident with vertices of the second end.

(4.13). *$P_1$ and $P_2$ are disjoint.*

If not, then the union of $P_1$ and $P_2$ contains a path $P_2'$ from $d_1$ to $w_2$ avoiding $A$ and $B$. $P_2'$ then satisfies the defining conditions for $P_2$, and so from (4.12) we deduce $d_1$ is in the second end of $D$, which is impossible because the ends of $D$ are disjoint (since $4\nu + 2\delta + 2 \geq 4$).

(4.14). *Conclusion.*

From (4.12) and (4.13) we deduce that $G$ has a rooted minor isomorphic to $P(\gamma - 1), (N - 4\nu) + (4\nu + 2\delta + 2 - 4)) = P(\gamma - 1, N + 2\delta - 2)$,[F14] as can be seen by deleting all vertices of $G$ except those in $A, B, D, P_1$ and $P_2$, and contracting the edges in $P_1, P_2$ and the ends of $D$. This contradicts the maximality of $N$, and completes the argument for Part 2.

[F14] **Misprint.** The brackets do not match. The tree meant is $P(\gamma - 1, (N - 4\nu) + (4\nu + 2\delta + 2 - 4))$, so the bracket after $\gamma - 1$ is removed.

## 5. Part 3: The Reduction of $P(\gamma, \delta)$ When $\gamma \geq 1$

We now assume that $1 \leq \gamma \leq \lambda$, and that for all $\delta \geq 2$, every rooted connected graph in $\mathscr{F}_\theta$ with no rooted minor isomorphic to $Q(\gamma, \delta)$ has path-width $\leq q(\gamma, \delta)$. We shall show, by induction on $\delta \geq 2$, that every rooted connected graph in $\mathscr{F}_\theta$ with no rooted minor isomorphic to $P(\gamma, \delta)$ has path-width $\leq p(\gamma, \delta)$.

Suppose then that $G \in \mathscr{F}_\theta$ is a rooted connected graph with no rooted minor isomorphic to $P(\gamma, \delta)$. If $G$ has no rooted minor isomorphic to $Q(\gamma, 4)$, then by hypothesis, its path-width is $\leq q(\gamma, 4) \leq p(\gamma, \delta)$ by (3.6)(iii). Thus we may assume that $G$ has a rooted minor isomorphic to $Q(\gamma, 4)$.

Now $P(\gamma, 2)$ is isomorphic to a rooted minor of $Q(\gamma, 4)$, and so $G$ has a rooted minor isomorphic to $P(\gamma, 2)$. It follows that $\delta > 2$. By induction on $\delta$, we have that every rooted connected graph in $\mathscr{F}_\theta$ with no rooted minor isomorphic to $P(\gamma, \delta - 1)$ has path-width $\leq p(\gamma, \delta - 1)$.

Let $H$ be $\mathbf{H}_\gamma$, rooted at its vertex of valency 2. Now $H$ is isomorphic to a rooted minor of $Q(\gamma, 4)$ and so $G$ has a rooted minor isomorphic to $H$. [p. 60] Choose an integer $N \geq 0$, maximum such that there exist $X_0, X_1, ..., X_N \subseteq V(G)$, disjoint, with the following properties:

(X1) $\rho(G) \in X_0$, *and* $G \,|\, X_0$ *rooted at* $\rho(G)$ *has a rooted minor isomorphic to* $H$;

(X2) *for* $1 \leq i \leq N$, $G \,|\, X_i$ *has a minor isomorphic to* $\mathbf{Y}_{\gamma-1}$.

Now all vertices of $\mathbf{Y}_{\gamma-1}$ have valency $\leq 3$, and so for $1 \leq i \leq N$, $G \,|\, X_i$ has a subgraph $B_i$ which is isomorphic to a subdivision of $\mathbf{Y}_{\gamma-1}$. Moreover, $G \,|\, X_0$ has a subgraph $B_0$ which consists of a subgraph $H'$ isomorphic to a subdivision of $H$, together with a path from $\rho(G)$ to the vertex of $H'$ corresponding to the root of $H$, such that the path has only this vertex in common with $H'$.

Clearly we may assume that $X_i = V(B_i)$ $(0 \leq i \leq N)$. Suppose that there exists $J \subseteq \{1, ..., N\}$ with $|J| = \delta \cdot 2^\gamma$, and disjoint paths $P_j (j \in J)$ of $G$, such that

(i) for each $j \in J$, $P_j$ has one end in $X_j$ and the other in $X_0$;
(ii) for each $j \in J$, $P_j$ has no interior vertex in $X_0 \cup \bigcup_{j' \in J} X_{j'}$.

Now $B_0$ has only $2^\gamma$ vertices distinct from $\rho(G)$ of valency 1, and $B_0$ is a tree; and so there are paths $Q_1, ..., Q_{2^\gamma}$ of $B_0$, each from $\rho(G)$ to some vertex of valency 1 distinct from $\rho(G)$, such that every vertex of $B_0$ is used by at least one of $Q_1, ..., Q_{2^\gamma}$. It follows that there exists $J' \subseteq J$ with $|J'| = \delta$ $(= 2^{-\gamma}|J|)$ such that for some $i$ $(1 \leq i \leq 2^\gamma)$, all the paths $P_j$ $(j \in J')$ have their terminal vertices on $Q_i$. But then $G$ has a rooted minor isomorphic

to $P(\gamma, \delta)$, as can be seen by deleting all vertices except those in $Q_i$ and $P_j$ and $B_j$ ($j \in J'$), and for each $j \in J'$ contracting all edges of $P_j$ except one. Thus there there is no such $J$.

By (2.9) (setting $V' = X_0$, $r = \delta \cdot 2^\gamma$, and $n = N$) we deduce that there exists $X \subseteq V(G)$ and $J \subseteq \{1, ..., N\}$ with $|X| + |J| \leq \Omega$, where $\Omega = \alpha^{\delta \cdot 2^\gamma}$ (and $\alpha = 2^{\theta^4}$, as before), such that $X$ separates $X_0$ from $X_j$ for every $j \in \{1, ..., N\} - J$. To complete the argument for Part 3, it suffices to show that every component of $G \backslash X$ has path-width $\leq \max(p(\gamma, \delta - 1), q(\gamma, 3\Omega + 6))$, by (3.6)(iv), (1.4), and (1.5). Thus, let $C$ be a component of $G \backslash X$. There are two cases.

(5.1). *If $X_0$ does not contain any vertex of $C$, then $C$ has path-width $\leq p(\gamma, \delta - 1)$.*

Let $P$ be a minimal path in $G$ from $V(C)$ to $X_0$. Then $P$ has at least one edge, and no interior vertex in $V(C) \cup X_0$. Let $v$ be the vertex of $C$ on $P$. Root $C$ at $v$. Suppose that $C$ has a rooted minor isomorphic to $P(\gamma, \delta - 1)$. Then $G$ has a rooted minor isomorphic to $P(\gamma, \delta)$, as can be seen by contracting all edges of $P$ except one and contracting one "half" of $B_0$, [p. 61] suitably chosen. This is a contradiction, and so $C$ has no such rooted minor. By induction, (5.1) is true.

(5.2). *If $X_0$ contains a vertex of $C$, then $C$ has path-width at most $q(\gamma, 3\Omega + 6)$.*

In this case $C$ contains no vertices of $\bigcup(X_j : j \in \{1, ..., N\} - J)$ since $X$ separates $X_0$ and this set. Let $P$ be a minimal path of $B_0$ from $V(C)$ to $\rho(G)$. Let $v$ be the vertex of $C$ on $P$. Root $C$ at $v$. Suppose that $C$ has a rooted minor isomorphic to $Q(\gamma, 3\Omega + 6)$. Then there exist $X'_0, ..., X'_{\Omega+1} \subseteq V(C)$, disjoint, such that $C \mid X'_0$, rooted at $v$, has a rooted minor isomorphic to $H$, and for $1 \leq i \leq \Omega + 1$, $C \mid X'_i$ has a minor isomorphic to $\mathbf{Y}_{\gamma-1}$. But then the sets

$$X'_0 \cup V(P), X'_1, ..., X'_{\Omega+1} \qquad \text{and} \qquad X_j (j \in \{1, ..., N\} - J)$$

satisfy (X1) and (X2), and there are $\Omega + 1 + N - |J| > N$ of them. This contradicts the maximality of $N$, and hence proves (5.2). This completes the argument for Part 3, and so proves (3.7).

## References


1. N. Robertson and P. D. Seymour, Graph minors. II. Algorithmic aspects of tree-width, submitted.

2. N. Robertson and P. D. Seymour, Graph minors. III. Planar tree-width, *J. Combinatorial Theory* (Ser. B), to appear.
3. N. Robertson and P. D. Seymour, Graph minors. IV. Tree-width and well-quasi-ordering, submitted.
4. N. Robertson and P. D. Seymour, Graph minors. V. Excluding a planar graph, submitted.